



BABA FARAH

SOFTWARE & EMBEDDED SYSTEMS ENGINEER

+212601461156

Farah.baba01@gmail.com

Tikiouine, AGADIR

<https://www.linkedin.com/in/farah-baba-367b7b30b/>

With a background in Computer Engineering and Embedded Systems, I combine expertise in hardware architectures and software development. Experienced in developing web applications, APIs, and connected solutions, I bring technical rigor, adaptability, and a strong problem-solving mindset to complex engineering challenges.

EDUCATION

- Master of Excellence in Computer Engineering and Embedded Systems, Ibn Zohr University – FSA, Agadir (2024–2026)
- Bachelor of Excellence in Computer Engineering and Embedded Systems, Ibn Zohr University – FSA, Agadir (2023–2024)
- Bachelor in Education – Computer Science, Higher School of Education and Training, Agadir (2020–2023)
- Baccalaureate in Physical Sciences (French option), Lycée Libres Agadir Ida Outanane (2022)
- Russian Language Diploma, Kharkiv Regional Institute of Public Administration, Ukraine (2020)
- Baccalaureate in Physical Sciences (Arabic option), Lycée Mohammed Derfoufi, Agadir (2019)

PROFESSIONAL EXPERIENCES

Embedded Systems Engineer — Final-Year Internship | OCP Jorf Lasfar | Feb. – Jun. 2026

- Designed and developed an embedded system for automated acquisition and processing of physical data.
- Contributed to system modeling, testing, and validation.
- Worked on the integration of hardware and software components to ensure system consistency and reliability.

Software Developer — Internship | MARSA MAROC, Agadir | Aug. – Sep. 2025

Developed a web application for SQL query management and execution using Laravel and MySQL.

Software Developer — Internship | National Ports Agency (ANP), Agadir | Jul. 2024

Developed a web application for IT asset management using Laravel and MySQL.

JAVA Developer — Final-Year Internship | ESEF Agadir | Feb. – May 2023

Developed an intelligent library management application in Java, including book, user, and loan management

RECENT TECHNICAL PROJECTS

HealthTech Hackathon – Context-Aware Cardio-Motor Anomaly Monitoring | Dec. 2025

Designed a smart wearable bracelet for the acquisition and monitoring of cardio-motor data.

Tools: ESP32, MAX30102, MPU6050, IoT, Machine Learning, Flutter, Firebase

Autonomous Mobile Surveillance Robot | Nov. 2025

- Designed an embedded robotic system for autonomous movement and environmental monitoring.
- Worked on the interaction between hardware and software components.
- Technologies: ESP32, ESP32-CAM, motion sensors, camera, motors.

Tools: Esp32, Esp32-CAM, motion sensors, camera, motors

AgriTech Hackathon – Plant Disease Detection | Dec. 2024

Developed an embedded IoT solution for plant disease detection based on environmental data and images.

Tools: Raspberry Pi 4, Camera, Soil Moisture Sensor, DHT22, Machine Learning, React

TECHNICAL SKILLS

Programming Languages: C, C++, Assembly, VHDL, Java, JavaScript, PHP, Python

Web & Mobile Development: HTML5, CSS, PHP, Java, Python, WordPress, React, Flutter, Flask

Databases: SQL, MySQL, SQL Server Management Studio, MongoDB

Big Data / ETL: Talend Open Studio, Hadoop, Spark, SSIS

Development Tools and Platforms: GitHub, Postman, Microsoft Office

Modeling and Simulation: UML, Merise, MATLAB, Quartus

Operating Systems: Windows, Linux, TCP/IP, Client-Server Architecture, Routing.

CERTIFICATIONS

- Technovate Hackathon AgriTech / HealthTech
- Automotive Engineering: Testing Strategies, UDS & Functional Safety (ISO 26262)

LANGUAGES

Arabic : Native ; **French** : Fluent ; **English** : Fluent ; **Russian** : Intermediate